

AI For Infrastructure & Cloud Architects

Training Duration: 18 hours

Learning Objectives & Target Audience

- Enable infrastructure and cloud architects to use Generative AI, Agentic AI, and AWS agents to design, automate, secure, and optimize cloud infrastructure.
- Participants will apply AI-driven workflows for RAG, agent orchestration, DevOps triage, security analysis, FinOps optimization, and capstone-based architecture scenarios.

Prerequisites

- Basic understanding of cloud infrastructure, AWS services, and AI/ML fundamentals

Contents		Duration (hours)
Module 1	Foundations & Paradigm Shift ○ Why is it important? (risks and failure scenarios) ○ LLM basics ○ Token economics ○ Prompt engineering ○ Retrieval-Augmented Generation (RAG) ○ Responsible AI	2
Module 2	Generative AI for Infrastructure ○ Amazon Bedrock ○ Amazon Q ○ Inferentia & Trainium ○ Agentic AI building blocks	2
Module 3	Strands ○ Concept and role in agent workflows ○ Orchestration of tasks and flows ○ Why Strands? ○ Benefits of Strands	2

Module 4	Agentic AI for Infrastructure ○ Bedrock Agent (basic) ○ Bedrock Knowledge Base (RAG) ○ Agent Core ○ Agents vs chatbots ○ Agent lifecycle ○ Agent architecture ○ Multi-agent systems ○ Agent observability (logs, metrics, traces)	4
Module 5	AWS DevOps Agent ○ DevOps agent lifecycle ○ Incident detection and triage ○ Root cause analysis ○ Telemetry inputs (logs, metrics, alerts) ○ Infrastructure actions (scale, restart, rollback)	4
Module 6	AWS Security Agent ○ Security agent lifecycle and configuration ○ Threat analysis and modelling ○ Code review ○ Security data sources (CloudTrail, findings, vulnerabilities) ○ Risk detection and remediation	2
Module 7	AWS FinOps Agent ○ FinOps agent lifecycle ○ Cost monitoring and anomaly detection ○ Resource utilization and right-sizing ○ Cost optimization strategies (savings plans, scheduling) ○ Forecasting and budgeting	2
Module 8	Capstone Project (expected to work using Design Thinking Approach)	5